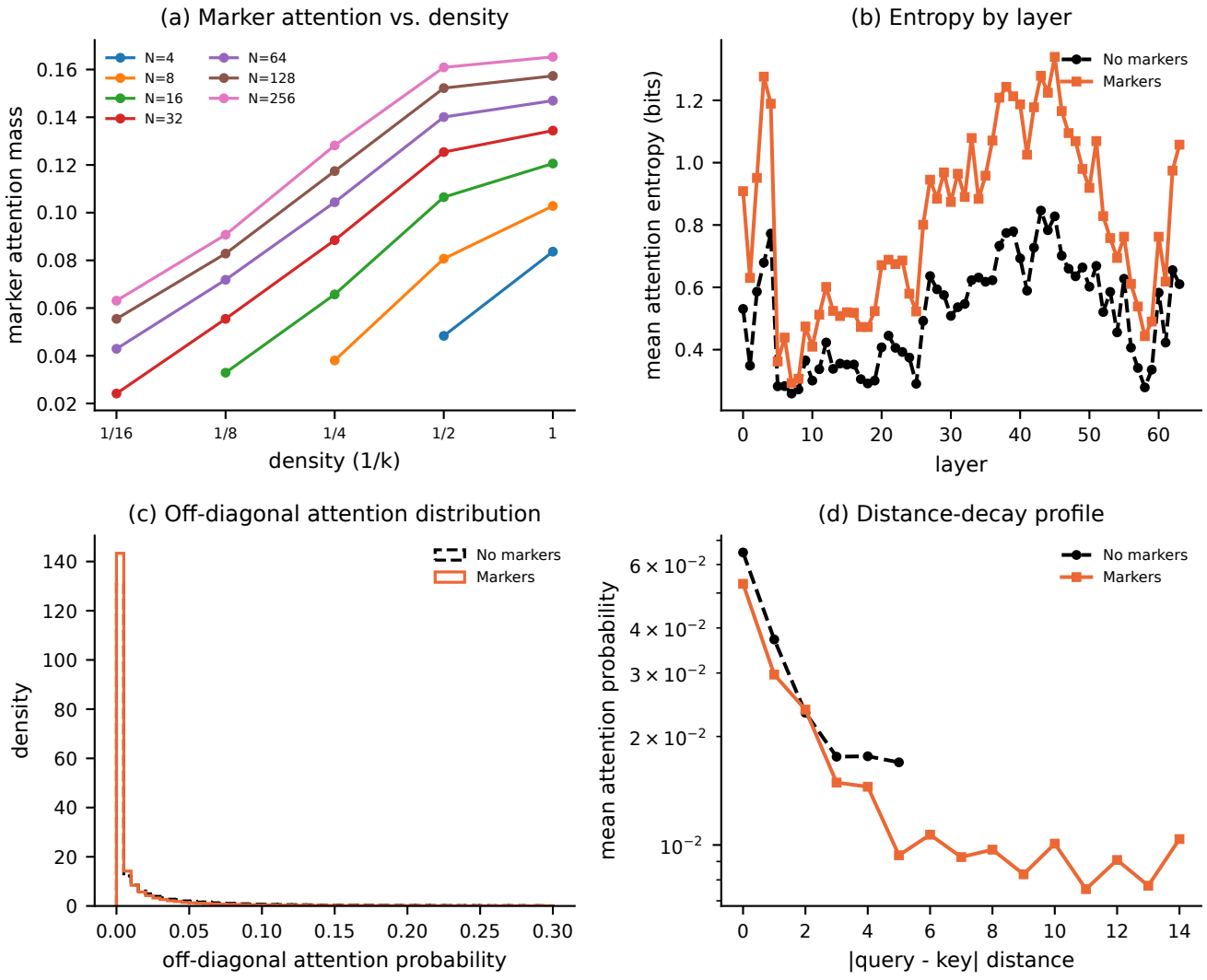

(a) Marker attention vs. density
marker attention mass
density (1/k)
N=4
N=8
N=16
N=32
N=64
N=128
N=256
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
1/16
1/8
1/4
1/2
1
(b) Entropy by layer
mean attention entropy (bits)
layer
No markers
Markers
0.4
0.6
0.8
1.0
1.2
0
10
20
30
40
50
60
(c) Off-diagonal attention distribution
density
off-diagonal attention probability
No markers
Markers
0
20
40
60
80
100
120
140
0.00
0.05
0.10
0.15
0.20
0.25
0.30
(d) Distance-decay profile
mean attention probability
|query - key| distance
No markers
Markers
$6 \times 10^{-2}$
$4 \times 10^{-2}$
$3 \times 10^{-2}$
$2 \times 10^{-2}$
$10^{-2}$
0
2
4
6
8
10
12
14